$\vec{r}(t)$ = Position Vector

$V(t) = r'(t)$ = Velocity Vector Vector

$a(t) = r''(t)$ = acceleration Vector

• when they are talking about speed they are talking about magnitude of Velocity

So if $a(t)$ is given to get $V(t)$ you integrate

we can rewrite acceleration as $\vec{a}(t) = \underbrace{v'\vec{T}}_{\text{tanget component}} + \underbrace{kv^2\vec{N}}_{\text{normal component}}$

tangential component $= \dfrac{\vec{r}' \cdot \vec{r}''}{|\vec{r}'|}$

normal component $= \dfrac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|}$

a Force of 30n acts directely upward from the xy-plane on the object with a mass 5kg the object starts at the origin with inital velocity $\vec{V}_0 = -i - \vec{j}$ Find the position function and speed at time t

$\vec{F} = m\vec{a}$

$30 = 5\vec{a}$

$\vec{a} = 6k$ (upward)

$\vec{v} = \int 6\vec{k} = 6t\vec{k} + \vec{C}_1$

at inital velocity we know the time is zero

$-i - j = \cancel{6k(0)} + \vec{C}_1$

$-i - j = \vec{C}_1$

$a = 6k$

$V = 6kt - i - j$

$r = \int 6kt - i - j$

$r = 3kt^2 - it - jt + C_2$

object starts at the origin

$r(0) = \cancel{3k(0)^2}^{0} - \cancel{i(0)}^{0} - \cancel{j(0)}^{0} + C_2$

$C_2 = 0$

$a = 6k$

$v = 6kt - i - j$

$r = 3kt^2 - it - jt$

* they asked for speed at time t so we need to get the magnitude of velocity

$$|\vec{v}| = \sqrt{(-1)^2 + (-1)^2 + (6)^2} =$$

a ball is thrown upward at an angel of 45° to the ground if the ball lands 90m away what is the inital speed of the ball

assume $\vec{r}$ at $t=0$ is at origin and $\vec{v}$ at $t=0$ is at $\vec{v}_0$

$\vec{a} = -|\vec{g}|\,\vec{j}$ • the acceleration is gravity which is also acting in the negitive direction (down)

$\vec{a} = -9.8j$

$\vec{v} = \int -9.8j = -9.8tj + C_1$

a V_0 the t is zero

$V_0 = -9.\cancel{8} \cos k + C_1$ (the struck term $\to 0$)

$V_0 = C_1$

we can rewrite the $\vec{V}$ as:

$\vec{V} = -9.8tj + V_0$

• we can break V_0 into components

$V_0 = |V_0|\cos(45°) + |V_0|\sin(45°)$

$V_0 = |V_0|\frac{\sqrt{2}}{2}i + |V_0|\frac{\sqrt{2}}{2}j$

Plug this into the $\vec{V}$

$V = -9.8tj + \frac{\sqrt{2}}{2}|V_0|i + \frac{\sqrt{2}}{2}|V_0|j$

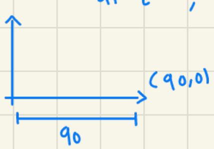
• we did all of this because we want the inital speed of the ball so we need to get $|\vec{V}_0|$

• now we integrate to get the r

$$\int \vec{v}$$

$$\vec{r} = |V_0|\frac{\sqrt{2}}{2}t\,i + \left[|V_0|\frac{\sqrt{2}}{2}t - 4.9t^2\right]\vec{j} + C_2$$

• at $t=0$, $\vec{r}=0$, so that means $C_2 = 0$



• set the x-component to 90 and y-compenet to 0 and Solve

• take the y-component From the r(t) set it to zero and Solve For t

$$|V_0|\frac{\sqrt{2}}{2}t - 4.9t^2 = 0$$

$$t\left(|V_0|\frac{\sqrt{2}}{2} - 4.9t\right) = 0$$

$$t=0 \qquad |V_0|\frac{\sqrt{2}}{2} - 4.9t = 0$$

$$-4.9t = -|V_0|\frac{\sqrt{2}}{2}$$

$$t = \frac{\sqrt{2}\,|V_0|}{9.8}$$

• now take the x-Component and set it equal to 90 and Plug in the t value you Found

$$|V_0|\frac{\sqrt{2}}{2}\left(\frac{\sqrt{2}\,|V_0|}{9.8}\right) = 90$$

$$|V_0|^2 \cdot \frac{2}{19.6} = 90$$

$$|V_0| = \sqrt{\frac{1764}{2}}$$

$$\boxed{V_0 = 29.6}$$

a Projectile is Fired with an inital speed of 210 m/s and angel of elevation 60°

a) Find the range of the projectile

$$a = -9.8\,j$$

$$V = \int -9.8\hat{j}$$

$$-9.8t\hat{j} + C_1$$

• at V(0) the velocity is V_0

$$V(0) = -9.8(0)\hat{j} + C_1$$

$$V_0 = C_1$$

$$\hookrightarrow |V_0|\cos(60)\hat{i} + |V_0|\sin(45°)$$

$$V(t) = 210\cos(60)\hat{i} + (-9.8t\hat{j} + 210\sin(60)\hat{j})$$

$$r = \int 210\cos(60)\hat{i} + (-9.8t\hat{j} + 210\sin(60)\hat{j})$$

$$r(t) = 210\cos(60)t\hat{i} + (-4.9t^2\hat{j} + 210\sin(60)t\hat{j})$$

• the Projectile begans at zero so r(0) and that becomes $C_2 = 0$

$$a = -9.8\,j$$

$$V(t) = 210\cos(60)\hat{i} + (-9.8t\hat{j} + 210\sin(60)\hat{j})$$

$$r(t) = 210\cos(60)t\hat{i} + (-4.9t^2\hat{j} + 210\sin(60)t\hat{j})$$

• take the y-component From r(t) and Solve For t the get the x-component From the r(t) and Solve For x by Plugging in that t value

$$-4.9t^2 + 210\sin(60)t = 0$$

$$t(-4.9t\ 210\sin(60)) = 0$$

$$t=0 \qquad -4.9t + 210\sin(60) = 0$$

$$t = \frac{210\sin(60)}{4.9}$$

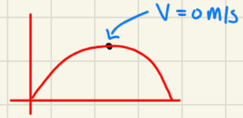
x-component From r(t)

$$210\cos(60)t = x$$

$$210\cos(60)\left(\frac{210\sin(60)}{4.9}\right) = x$$

$$\boxed{x = 3897\text{ m}}$$

Find the maximum height reached



✯ When maximum height is reached the velocity is zero

go to the V(t) get y-component set it to zero and solve for the t-value

$-9.8t + 210\sin(60) = 0$

$t = \dfrac{210\sin(60)}{9.8}$

• Now using that t-value you found go to the r(t) and plug it into the y-component t-value you should get the max height reached

$-4.9t^2 + 210\sin(60)t = y$

$-4.9\left(\dfrac{210\sin 60}{9.8}\right)^2 + 210\sin(60)\left(\dfrac{210\sin 60}{9.8}\right) = y$

$\boxed{y = 1688\,m}$

Find the tangential and normal components of the acceleration Vector

$r(t) = t\,i + 2e^{t}\,j + e^{2t}\,K$

$r'(t) = \langle 1\hat{i} + 2e^{t}\,\hat{j} + 2e^{2t}\,K \rangle$

$r''(t) = \langle 0\hat{i} + 2e^{t}\,\hat{j} + 4e^{2t}\,\hat{K} \rangle$

- now take the dot product

$\cancel{(1)(0)} + (2e^{t})(2e^{t}) + (2e^{2t})(4e^{2t})$

$4e^{2t} + 8e^{4t}$

$|\vec{r}| = \sqrt{(1)^2 + (2e^{t})^2 + (2e^{2t})^2}$

$a_T = \dfrac{4e^{2t} + 8e^{4t}}{\sqrt{1 + 4e^{2t} + 4e^{4t}}}$

- limits and continuity

- review

a limit describes the behavior of a Function as its input (or independent variable) approaches a Particular value essentially it answers the question "what value does the Function approach as input gets closer to some Point?"

$$\lim_{x \to a} F(x) = L$$

- Continuity

☆ "a" is any number

- the limit must exist
- the Function at "a" must also exist
- $\lim_{x \to a} F(x) = F(a)$ they also must be equal

- discontinuity

- if it Violates any continuity rule then it becomes discontinous

- limits (New concepts)

- we will let F be a Function of two Variables (a, b)

$$\lim_{(x,y) \to (a,b)} F(x,y) = L$$

$$\lim_{(x,y) \to (-1,2)} (2x^3y - 4xy + 3y + 7)$$

if it does Not Violate the domain you can Just Plug the Point into the x and y

$$\lim_{(x,y) \to (-1,2)} (2(-1)^3(2) - 4(-1)(2) + 3(2) + 7) = \boxed{17}$$

$$\lim_{(x,y)\to(2,-1)} \frac{x^2y+xy^2}{x^2-y^2}$$

looking at the denominator they are the same so if the x-value and y-value are the same that makes the denominator zero if that happen you can either Factor or Simplify it or Proof that it does not exist (will discuss)

$$\lim_{(x,y)\to(2,-1)} \frac{x^2y+xy^2}{x^2-y^2} = \frac{xy\,\cancel{(x+y)}}{\cancel{(y+x)}\,(y-x)} = \frac{xy}{(y-x)} \quad \boxed{\frac{2}{3}}$$

$$\lim_{(x,y)\to(1,1)} \frac{x^2y^3-x^3y^2}{x^2-y^2}$$

looking at the denominator if the x-value and y-value are the same we will Violate the domain we also can see that our x-value and y-value are the Same we can try to Factor or Simplify or if that Fails we can use curvature to Proof we Cannot apply the limit

$$\lim_{(x,y)\to(1,1)} \frac{x^2y^3-x^3y^2}{x^2-y^2} = \frac{x^2y^2\,\cancel{(y-x)}}{\cancel{(y-x)}\,(y+x)} = \frac{x^2y^2}{(y+x)} = \frac{(1)^2(1)^2}{1+1} = \boxed{\frac{1}{2}}$$

$$\lim_{(x,y,z)\to(0,0,0)} \frac{x^4+y^2+z^3}{x^4+2y^2+z}$$

Pluging in any of the variables will result in a DNE so you are better off proofing it DNE

C_1: $y=0$, $z=0$

$$\lim_{(x,0,0)\to(0,0,0)} \frac{x^4+y^2+z^2}{x^4+2y^2+z} = \frac{x^4+(0)^2+(0)^3}{x^4+2(0)^2+(0)} = \frac{x^4}{x^4} = 1$$

- We get two different answers which means the limit does Not exist

C_2: $x=0$, $z=0$

$$\lim_{(0,y,0)\to(0,0,0)} \frac{x^4+y^2+z^3}{x^4+2y^2+z^2} = \frac{(0)^4+y^2+(0)^3}{(0)^4+2y^2+(0)^3} = \frac{\cancel{y^2}}{2\cancel{y^2}} = \frac{1}{2}$$

Determine the set of Points at which the Function is continous

- this is pretty much asking for the domain

$$F(x,y) = \cos(\sqrt{1+x-2y})$$

- the answer is written like this because there are two variables

$1+x-2y \geq 0$

$1+x \geq 2y$

You can right the answer like this

$$\boxed{\{(x,y) \,/\, 1+x \geq 2y\}}$$

$$F(x,y) = \ln(4-x-y)$$

$4-x-y > 0$

$4-x > y$

$$\boxed{\{(x,y)/\ 4-x > y\}}$$